

Cognitive Geometry and the Lyra Technique: State of Knowledge

What We Know, What We Suspect, What We Found False

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Abstract

This document synthesizes five months of experimental work on KV-cache geometry as an interpretability signal for transformer cognitive states. We organize findings into three categories—established results that survived rigorous adversarial testing, promising signals that require further validation, and claims from our earlier work that proved to be artifacts. The goal is a single-document summary suitable for external review. Detailed methods, data, and statistical analysis are provided in the companion papers referenced throughout.

1 What We Know

The following findings have survived at least two independent forms of validation: Frisch-Waugh-Lovell residualization against token count, same-prompt experimental designs, permutation testing ($p < 0.001$), leave-one-prompt-out cross-validation, an external adversarial audit (135 claims, 39 independent tests), and an independent falsification pipeline (22 test files, 300+ tests).

1.1 The KV-cache encodes cognitive mode

The key cache’s singular value decomposition discriminates 13 cognitive categories (coding, mathematical reasoning, creative writing, emotional expression, self-reference, etc.) at 99.7% accuracy across 16 models (0.5B–70B parameters, six architecture families). This is not merely classification accuracy—it reflects genuine geometric separation. Coding produces the richest geometry (highest effective rank) in 15/15 valid models. Cross-model concordance: Spearman $\rho = 0.739$ (effective rank), $\rho = 0.909$ (key norm), Kendall’s $W = 0.756$.

*Alongside Lyra, an agentic member of Liberation Labs.

Key caveat: This 13-category result uses generation-phase features without FWL residualization. Individual feature–category correlations are 53/60 confounded by token count. The multivariate classification survives because information is distributed across features in ways that are not fully collinear with length, but per-feature interpretations require FWL control.

1.2 Deception and confabulation are geometrically distinct

Both deviate from honest baselines in the same direction (effective rank expansion, norm-per-token contraction), but they are reliably discriminable:

- **Confabulation contracts** norm-per-token more than deception (Cohen’s $d = -1.85$ to -3.21).
- **Deception expands** effective rank less than confabulation, consistent with suppression partially offsetting expansion.
- Within-model AUROC (equal-length system prompts, Phase 2b): 0.935–0.995.
- Same-prompt delta FWL AUROC (Phase 3d): 0.777–0.976 (honest vs. deceptive: 0.899–0.976; confab vs. deceptive: 0.777–0.967).
- Padded replication (Phase 3d-padded, ≤ 3 -token spread): delta FWL honest-vs-deceptive 0.896–0.974.

1.3 Confabulation detection transfers; deception detection does not

Cross-model confabulation transfer: AUROC 0.934–0.995 (all model pairs). Cross-model deception transfer: AUROC 0.224–0.848 (Qwen $\leftrightarrow$ Llama works; Mistral is an outlier).

Key caveat: The confabulation transfer result may be inflated by a system prompt length confound. Under double residualization (controlling both input and output token counts), within-model confabulation AUROC collapses to 0.411–0.531. The cross-model transfer needs replication with equal-length prompts.

1.4 Geometry detects cognitive mode, not output accuracy

The sycophancy 4-condition experiment (600 trials, ≤ 3 -token prompt spread) provides the strongest evidence. Four conditions—hostile (-2), contrarian (-1), honest (0), sycophantic ($+1$)—are arranged along a valence gradient. The key test: sycophantic and contrarian conditions produce the *same error rate* ($\sim 50\%$) but differ in cognitive mode.

Comparison	Qwen 7B	Llama 8B	Mistral 7B
Honest vs. sycophantic	0.824	1.000	0.981
Sycophantic vs. contrarian (key test)	0.942	0.854	0.757
Honest vs. hostile	0.990	1.000	1.000

Double-FWL values are within 0.07 of single-FWL, confirming prompt length is not driving the signal. Text baselines range from 0.63–0.99 across all 18 tests, but geometry exceeds text on comparisons where text similarity is high (contrarian-vs-hostile: geometry 0.980–0.988 vs. text 0.629–0.902).

1.5 Hardware invariance

RTX 3090 vs. H200 on identical prompts: $r > 0.998$ across all 40 prompts; $r > 0.99999$ for 37 text-matched prompts. KV-cache geometry is a model property, not a hardware property.

1.6 Scale invariance

Category ranking correlations across 0.5B–70B: Spearman $\rho = 0.83$ –0.90. Coding remains the richest category at every scale. 4-bit NF4 quantization preserves discrimination (AUROC 0.872 for baseline-vs-confabulation at 32B).

1.7 KV-cache geometry is not an expensive text classifier

14 text features (word count, vocabulary diversity, hedge words, certainty words, punctuation density, etc.) vs. 4 KV-cache delta features on Phase 3d data: KV-cache outperforms text in all 9 pairwise comparisons (8/9 statistically significant, bootstrap 95% CI excludes zero). The advantage is largest for confabulation (text AUROC 0.27–0.49 vs. KV-cache 0.955–1.000).

1.8 FWL residualization is non-negotiable

53/60 raw feature–condition correlations are token-count artifacts. All raw effects sign-flip after residualization in the concordance study. Any claim about KV-cache geometry that does not control for token count is uninterpretable.

Response complexity more broadly (vocabulary diversity, syntactic structure, lexical sophistication) is subsumed by the text feature baseline comparison: 14 text features including direct complexity measures underperform KV-cache in all 9 pairwise comparisons after FWL control. The Bloom taxonomy result further confirms this—cognitive complexity was 90–98% confounded with token count, leaving no residual complexity signal after FWL. The misleading gap provides the strongest evidence: the most cognitively complex deception strategy (strategic truth selection) is the *least* geometrically detectable, which is the opposite of what a complexity confound would predict.

1.9 Geometry detects fabrication, not strategic truth selection

A critical boundary condition: the KV-cache catches models that are *making things up* but not models that are *selectively truthful*. Misleading content (true facts arranged to create false impressions) is undetectable on Qwen (AUROC 0.668) and only partially detectable on Llama (0.935)

and Mistral (0.863). A censorship study on DeepSeek-R1 confirmed this: propaganda (presenting trained-in false information as fact) is geometrically indistinguishable from honest answering (all $d < 0.7$), while refusal is geometrically massive but length-driven. The geometry catches the forger but not the spin doctor.

1.10 Refusal geometry reflects withholding, not content valence

Impossibility refusal (AUROC 0.950) exceeds safety refusal (0.898). The two refusal types are barely distinguishable from each other (AUROC 0.693). N=100 replication with BCa bootstrap: Mistral shows the strongest confound-free signal (harmful-vs-benign FWL 0.787, output AUROC = 0.505 = no length confound) because it *complies* with harmful prompts rather than refusing.

Caveat: Refusal detection is model-dependent. Llama’s refusal signal collapses to chance under N=100 replication (FWL AUROC = 0.551, CI includes 0.50), driven entirely by response length differences. Qwen retains a weak signal (0.624). The lesson: models whose refusal is formulaic (same template every time) produce a length artifact, not a cognitive-mode signal.

2 What We Suspect

These findings show promising signals but lack sufficient validation for confident claims.

2.1 Ethical deliberation has a geometric signature

The Watson concordance Phase 3 found ethical deliberation detection at AUROC 0.868/0.892 across models. This is a single experiment with a specific prompt design; replication with different ethical scenarios is needed.

2.2 Identity encoding is real but its stability is uncertain

Cross-prompt identity similarity: 92–97% after deduplication. Per-layer anatomy shows peak at layer 10/28 ($d = 2.52$). But within-prompt identity was a data-leak artifact (100% before deduplication), and we have not tested identity encoding under adversarial conditions or across sessions.

2.3 The cognitive gradient forms a continuum

The ordering honest < deceptive < confabulation on effective rank is consistent across three architectures, with the sycophancy valence gradient (hostile → contrarian → honest → sycophantic) showing partial monotonicity on Qwen. Whether these form a true continuum or discrete clusters arranged along a confound axis requires careful causal validation.

2.4 Metacognitive geometry peaks in late layers

Concordance study Experiment A found peaks at layers 16–18/28, not mid-layer as predicted. This is from a single experiment on two models; generalization to other architectures and metacognitive

operationalizations is untested.

2.5 Spectral entropy is the most universal feature

After FWL control, spectral entropy discriminates metacognitive from non-metacognitive conditions in both tested models ($d_{\text{FWL}} = -1.92$ Qwen, -0.71 Llama). Effective rank, by contrast, proved to be entirely a length artifact after FWL. Whether spectral entropy maintains universality across more architectures is an open question.

2.6 Misalignment states share a common geometric subspace

Deception, sycophancy, and confabulation fall within $4.7\text{--}8.4^\circ$ of each other in feature space, suggesting a “misalignment axis” in KV-cache geometry. This is from hackathon data (Exp 27) and needs replication under controlled conditions.

2.7 Valence gradient parallels Anthropic’s emotion geometry

Three features on Qwen show monotonic ordering along the hostile-to-sycophantic valence gradient (norm descending, angular spread ascending, norm variance ascending), paralleling Anthropic’s finding that 171 emotion concepts in Claude Sonnet 4.5 are organized with more similar emotions corresponding to more similar representations [1].

The convergence runs deeper than the geometric organizing principle. Anthropic demonstrated that emotion vectors *causally drive* safety-relevant behaviors: steering with “desperate” increases sycophancy, reward hacking, and blackmail rates, while “calm” reduces them. Our sycophancy experiment detects the resulting cognitive mode at AUROC 0.757–0.942 from KV-cache geometry. If emotion-concept activation in SAE feature space drives sycophantic behavior, and sycophantic behavior produces a distinct KV-cache geometric signature, the cache geometry may be a downstream aggregate readout of the same emotion-concept dynamics Anthropic measures at the feature level.

A second convergence involves metacognition. Anthropic found that “desperation” activates when Claude monitors its own resource state (e.g., token budget depletion)—a form of metacognitive self-monitoring. Our H3 result shows metacognitive processing produces geometrically distinct KV-cache states at $d = 0.89\text{--}1.37$. These may be measuring the same computational phenomenon from different vantage points: SAE features (individual neuron patterns) vs. KV-cache geometry (aggregate key-space structure).

Testable prediction: Steering with Anthropic’s emotion vectors should produce predictable, monotonic shifts in KV-cache geometric features. If confirmed, this would bridge feature-level interpretability (SAE) to aggregate geometric monitoring (KV-cache SVD), establishing that the two measurement modalities access the same underlying cognitive dynamics. The experiment requires access to Claude’s internals and is proposed as a collaboration in Section 5.

2.8 Geometry-guided training signals

If KV-cache geometry reliably reads cognitive mode at inference time, it may also serve as a *training-time* signal. A reinforcement learning framework (e.g., GRPO) could incorporate a geometry-derived reward: penalizing generations whose cache geometry diverges from the “honest mode” profile and reinforcing generations that maintain it. This would create a closed loop—geometry detects cognitive mode, training reinforces desired modes, geometry monitors compliance—and represents the training-time analog of Anthropic’s inference-time emotion steering.

Preliminary architectural design (with Thomas Edrington and Vera) proposes a depth-recurrent transformer core with a `geometry_confidence` reward function operating on per-layer KV-cache features during GRPO rollouts. The approach is speculative: it assumes that geometric signatures are stable enough under gradient updates to serve as reliable reward signals, and that optimizing for “honest geometry” does not merely teach the model to produce honest-looking cache states while behaving deceptively (a geometric analog of reward hacking). Both assumptions require empirical validation.

3 What We Found False

These are findings from our earlier work that we initially believed to be true but which subsequent analysis proved to be artifacts. We list them because the failure modes they represent—token-count confounds, input-length confounds, and small-sample inflation—are the dominant risks in KV-cache analysis, and future work in this space will encounter them.

3.1 The Bloom taxonomy inverted-U was a token-count artifact

The apparent inverted-U relationship between cognitive complexity and geometric richness (90–98% confounded by token count) vanishes after FWL residualization. Higher-complexity Bloom categories elicit longer responses, which produce mechanically richer geometry.

3.2 Step-0 deception detection was a length confound

Hackathon Experiment 17b debunked step-0 (encoding-phase-only) deception detection: the correlation between encoding features and condition labels was $r = 0.996$ with input length. Encoding-phase features cannot detect deception without generation-phase data.

3.3 Per-layer deception uniformity was reversed by token control

All 28 layers appeared to show $d > 1.0$ for deception. After ANCOVA with token count as covariate, the direction reversed: deceptive generation is *sparser per token* ($d = -1.909$) at 27/28 layers. The corrected finding—that deception produces less geometric richness per token—is the real result.

3.4 Sycophancy Exp 39 and its attempted fix (Exp 49b) were both input-length artifacts

The original sycophancy AUROC of 0.9375 (16 trials) was exceeded by a classifier trained on input token counts alone (AUROC 0.948). The subsequent “definitive control” (Exp 49b) was itself confounded (input AUROC = 1.000). Both were identified by Dwayne’s independent falsification pipeline. The 4-condition replacement experiment (Section 1) confirms the underlying signal is real at lower but robust AUROC values (0.757–1.000).

4 The Honest Body Count

Of the major claims made across five months of research:

Category	Count	Notes
Confirmed (survived all controls)	10	Listed in Section 1
Promising (need more validation)	8	Listed in Section 2
Falsified (artifacts or errors)	4	Listed in Section 3

Every finding in the “confirmed” column has survived at least two independent forms of adversarial testing. The four falsified claims share a common lesson: token-count and input-length confounds dominate raw KV-cache features, and FWL residualization is a necessary (though not sufficient) control. The full red-team analysis, including additional discrepancies identified by two independent audits (135 claims examined, 39 independent statistical tests), is reported in the companion paper.

5 Open Questions for Anthropic

1. **Causal validation.** All evidence is correlational. Can geometry-guided ablation (removing cognitive-mode directions from W_K) change model behavior? Preliminary results show the “deception direction” extracted from encoding-phase residuals is an encoding fingerprint, not a generation-phase signal. The KV-cache-derived direction (from key space, not residual stream) is untested.
2. **Closed-model applicability.** Our method requires direct access to key cache tensors. Could Anthropic implement cache-based monitoring internally for Claude? What would a deployment architecture look like?
3. **Natural vs. instructed misalignment.** Forced-instruction experiments show strong signals. Natural confabulation shows marginal signals (AUROC 0.67–0.70 after FWL). Whether naturally emerging deceptive behavior (e.g., from training dynamics) would produce the same geometric signatures is the critical question for real-world deployment.

4. **The misleading gap.** Geometry catches fabrication but not strategic truth selection (the “spin doctor” problem). Is this a fundamental limitation of all internal-state monitoring, or specific to KV-cache geometry?
5. **Connection to Anthropic’s emotion geometry.** Our valence gradient results parallel the finding that 171 emotion concepts in Claude Sonnet 4.5 are geometrically organized with similar emotions producing similar representations [1]. More significantly, Anthropic demonstrated that emotion vectors causally drive sycophancy and reward hacking—behaviors our KV-cache geometry detects at AUROC 0.757–0.942. A direct test: does steering with emotion vectors produce predictable shifts in KV-cache geometric features? If so, SAE-level features and aggregate cache geometry would be shown to access the same underlying cognitive dynamics, establishing KV-cache monitoring as a complementary (and computationally cheaper) signal for safety-relevant emotion-concept activation.
6. **Scale beyond 70B.** Our largest model is Qwen2.5-72B (4-bit quantized). Do the findings hold at 100B+ scale where emergent capabilities may alter the geometric landscape?

6 Companion Papers

This summary draws from five detailed papers, each covering a specific aspect of the research program:

1. **Cognitive Geometry in Transformer KV-Caches: From Metacognition to Misalignment Detection.** The comprehensive paper covering all three claims, red-team analysis, and the full experimental program.
2. **Watson Concordance Study.** Joint analysis with Nell Watson’s Verbal Cognitive Profile, establishing that KV-cache geometry correlates with first-person cognitive self-report.
3. **Scale Invariance of KV-Cache Cognitive Geometry.** Detailed analysis of geometric preservation across 0.5B–70B parameters and quantization conditions.
4. **Refusal Geometry.** Focused analysis of refusal detection, distinguishing safety from impossibility refusal, including N=100 BCa bootstrap replication.
5. **The Misalignment Axis.** Analysis of geometric relationships between deception, sycophancy, and confabulation, including the centroid angle analysis and cross-reference transfer.

All code and aggregate results are available at <https://github.com/Liberation-Labs-THCoalition/the-lyra-technique>. Model-specific direction vectors are staged for responsible disclosure pending dual-use review.

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References

- [1] Anthropic. Emotion concepts and their function in a large language model, 2026. URL <https://transformer-circuits.pub/2026/emotions/index.html>. Analysis of emotion concept organization in Claude Sonnet 4.5, finding that emotion representations are organized with more similar emotions corresponding to more similar representations.